

Open Work

The single list. Everything still open, on every platform, re-read in the code against the latest `dev`. Each item says what happens today in plain words, what the product must do instead, and where the work is. Item numbers are stable across revisions.

Rev 6 · 21 August 2026

Read against `dev` : backend `00c6e35` · web apps `cb2d905d` · mobile `96aed6a6` · landing `bb99528`

HOW THIS LIST WORKS

One item, one place

This is the only open-work list. When a batch is handed to a developer it comes *out* of here and lives in the sheet they were given, so nothing is ever described in two documents at once. One batch is out right now: *Web — speed cleanup* (17 August), which is why the web performance items are not below.

Product decisions — new statuses, which buttons appear, how a workflow changes — are made in the item, not left to the developer. Only the technical how is theirs. Anything that still needs an answer from you is in **Decisions** at the back and should not become a ticket until it is answered.

SINCE 12 AUGUST

15 items closed on dev, 8 more written and waiting to be pushed

The head backward threshold is now -5° everywhere — risk verdict, Word report, RAMP and phone all agree, trunk stays -10° . That closes DEC-1 without you having to answer it.

A further eight calculation and report items are written but not yet on `dev`, so they are not in the list below either. Both sets are listed at the back.

Where the work is

The same items, indexed by platform, so a batch can be pulled together without splitting this document. Numbers only — the items themselves are in the sections that follow.

PLATFORM	COUNT	ITEMS
Backend	20 items	A5 · B2 · B6 · B7 · B9 · C1 · C2 · C3 · C4 · C5 · C6 · E2 · E8 · E9 · E10 · E13 · F7 · G5 · H7 · H8
Web	24 items	A1 · A2 · A11 · A12 · E1 · E3 · E4 · E6 · E7 · E11 · G1 · G2 · G3 · G4 · G6 · G7 · H1 · H2 · H3 · H4 · H5 · H6 · H12 · H13 · H14 · H15 · H16 · H17 · H18
Mobile	18 items	A7 · D1 · D2 · D4 · D5 · D7 · D8 · D9 · D10 · E6 · E12 · F10 · F11 · F12 · H8 · H9 · H10 · H11
With the web team	—	F1 to F6, F8, F9 — handed out 17 August as <i>Web — speed cleanup</i> . Not repeated here.

SUGGESTED FIRST BATCHES

If you hand work out now

Web: G1 with G2, G6 and G7 — the pipeline gate. Until that exists, anything fixed on the web side can be silently undone again. Then G3 (tests on the scoring), then H2 with H3, H4 and H6 (delete the v1 screens), then E1 with E3, E4 and E11 (one way to show an error, which needs E2 agreed with the backend first).

Mobile: D1 with D2, D7 and D8 — finish the move onto the Work Library API. They are the same code and there is a deadline attached: the old endpoints still answer, and the phone stops working the day they do not. Then D5 and D10, then the tolerance items D4 and H11.

Backend: E2 first, because E1 and E3 on the web are built on top of it.

A · Broken for users right now

A1 A measurement you retry from Home never comes back HIGH WEB

The Home list shows measurements that need attention. When you retry a failed one, it disappears from the list immediately — and the app remembers that it is hidden, in your browser, forever.

So if the retry then fails again, the measurement never reappears on Home. You will not be told and you will not see it there again. It is also hidden only on your computer, so a colleague still sees it as if nothing happened, and if you sign in from a different machine it is back.

DO

An item should leave the Home list because the thing that put it there is resolved, not because someone clicked something. Work that out on the server, not in the browser.

A2 A manual RAMP you saved cannot be opened again HIGH WEB

Create a RAMP by hand and you are taken straight into the form. You can fill part of it and press Save. But once you go back to the assessments list, that assessment cannot be opened — clicking the row does nothing.

Your answers are saved and unreachable, so the only way forward is to delete it and start over.

DO

An unfinished manual RAMP must reopen its form when you click it, the same way RULA and REBA already do.

NOTE FOR YOU

The code says this out loud — the comment above the click handler reads “pure-manual RAMP have no click-through”. It was written that way on purpose and never finished. Design it together with the save-and-resume feature at the back; it is the same screen.

A5 Searching for a worker still misses part of an ID BACKEND

Half of this improved on its own. The Workers search now matches the start of any *word* in the ID, so typing `0451` finds `SE2026-0451`. What still fails is an ID with no separators: in `SE20260451`, only `SE` matches.

DO

Match anywhere in the ID. While you are there: age accepts any number at all, so ages like 333 are in the data. Limit it to something sensible.

A7 Picking an activity on the phone looks like it worked when it did not HIGH MOBILE

Choosing an activity closes the picker whether or not the choice was saved. The failure is reported to Sentry and the screen closes anyway, so the user goes back to the recording screen believing they picked something. The conditions screen behaves the same way.

DO

Keep the screen open when the save fails and show the problem there until it succeeds or the user backs out deliberately. Closing the screen must mean it saved.

A11 The admin panel throws away what you typed WEB

A failed save closes the organisation form and loses everything entered. Filters break on names containing “&” or spaces. The organisation filter silently drops selections the user did not touch. And the counts do not refresh after adding a user.

NOTE FOR YOU

Carried over from 10 August and not re-read since. No commit has touched those screens, so nothing can have changed — but nobody has opened them either.

A12

A recording of zero seconds is treated as a real measurement

WEB

BACKEND

A recording that captured nothing is marked complete, sits in the list like any other, and can be sent for a risk assessment. Nothing in the code looks at the duration before doing any of that.

DO

Mark it Failed instead, and do not offer it for assessment.

B · The numbers we report are wrong

Only what is not already written. Hand distance, the AI section, the 1.5 colour rule, the head column and the empty MEC questions are all fixed and waiting to be pushed — they are listed at the back.

B2 The head's own angle is never reported anywhere BACKEND

MEC questions 11.1 and 11.2 — the screen calls them “neck bending” — report the difference between the head and the trunk. The backend's own internal name for them says they should be the head's own angle. The consequence either way is that the head's own angle appears nowhere in MEC, so a worker whose head moves with their body always scores near zero on question 11.

NOTE FOR YOU

Waits on DEC-6. With 12.1 and 12.2 being deleted, the question is simply what 11 is supposed to measure. Nothing should be built until that is answered.

ID WHERE TASK

- | ID | WHERE | TASK |
|----|---------|--|
| B6 | Backend | In MEC question 11 the head and trunk readings are matched by their position in the list instead of by time. The two sensors record at different rates, so from the first dropped reading onwards it compares the head at one moment against the trunk at another, and the leftover readings are silently discarded. Everywhere else in the system pairs by time. Ask why the earlier attempt at this was reverted before redoing it. Also waits on DEC-6. |
| B7 | Backend | The total time reported for a limb can be longer than the recording itself. Every raw read is bounded by the session window plus a deliberate one-day margin on each side, so a limb total can count readings from up to a day either side of the recording. |
| B9 | Backend | Six checkboxes in Generate report do nothing. Remove four of them — Worker details, Recorder details, Session duration, Session metadata — the report makes no sense without those tables, so they should not be optional. Make the other two, AI risk summary and Appendix, actually include or exclude their section. |

C · Security — nothing is broken, but the server trusts the app

READ BEFORE ASSIGNING

None of these affect a user, and none will show up in testing

Our apps always send the right values, so the product works. The gap is that signing in proves who someone is, not what they are allowed to touch — so someone calling the API directly, outside our apps, can send something ours never would. There is no sign any of it has been used. Whether to fund it now is DEC-3.

C1 The server decides whose data to return from the request, not from who is signed

in HIGH BACKEND

Several Work Library endpoints — activities, conditions, and the labels put on a recording — pick which company's data to use from a number in the web address rather than from the signed-in user. Any valid login can read, create, edit or delete another company's data by sending a different number.

DO

Take the company from the signed-in user and ignore the one in the address.

NOTE FOR YOU

Six endpoints still work this way; four in the same area were already corrected, so there is a pattern to copy. This is not dead legacy code — those rows are today's activities and conditions, and both apps use them.

ID	WHERE	TASK
C2	Backend	A recording's activity or condition can be re-pointed at another company's. Attaching one is checked; changing it afterwards is not. Apply the same check on edit.
C3	Backend	The default for the whole API is still "anyone may call this", so any endpoint where a developer forgets to say otherwise is public. Make the default "must be signed in" and mark the genuinely public ones — sign-in, register, password reset.
C4	Backend	The backend answers to any address, and the full list of its endpoints is readable without signing in. Restrict both outside development.
C5	Backend	The statistics export answers differently for a recording that belongs to another company than for one that does not exist, which lets an outsider work out which recording numbers are real. Answer the same way in both cases.
C6	Backend	Sign-in treats any unexpected failure as "not signed in", so a database outage would quietly make every request anonymous instead of reporting a problem.

D · The phone and the backend disagree

Things the app sends that the backend does not accept. Each one fails quietly. D1, D2, D7 and D8 are the same code — do them as one job.

D1 The phone still calls the old address for labels, categories and events HIGH MOBILE

The backend moved activities and conditions to the Work Library and the web app followed. The phone still calls the old `/categories/...` address for all four of its calls: fetching a category with its labels, creating a category, creating a label, and uploading the session's events.

This works today only because the old endpoints have not been switched off. The day they are, every activity and label action on the phone stops, and no session events upload at all.

DO

Move all four calls onto the Work Library API, the one the web app uses. Statuses and the activity lists are already on the new API, so this is a finishing job rather than a rewrite — follow what is there.

Do this before D2, D7 and D8. They touch the same code, and doing them separately means touching it three more times.

D2 Break markers never save, and they take the rest of the session with them HIGH MOBILE

A break is recorded against label number 0. That label does not exist on the backend, so the upload is refused. Because all of a session's events go up in a single request, one break means *none* of that session's events are saved — not the activities, not the tasks, nothing.

DO

A break is a real thing that happened during the recording and it must reach the server. Send it as something the backend accepts rather than as a label that does not exist.

Whatever the shape, one event failing must never stop the others: send them so a single bad event is rejected on its own and the rest still land.

NOTE FOR YOU

Check what actually reaches the server, not what the app thinks it sent. This has looked fine from the phone side for a long time.

ID WHERE TASK

D4	Mobile	A work-cycle step is allowed to have no duration — for instance one imported from a spreadsheet with the time column blank. The phone requires a number, so the whole step list fails to load and the operator cannot start recording at all. Treat a missing duration as no duration, not as an error.
D5	Mobile	Per-step timing is thrown away for the whole session. The phone labels the recording type slightly differently from what the backend matches on, so no step events are recognised and a work-cycle recording ends up with no per-task numbers at all. Check the backend's own list of accepted values rather than guessing from the field name.
D7	Mobile	A category that sits inside another category crashes the labels screen. It only works today because none of the categories in use have a parent, and nothing prevents one being created.
D8	Mobile	The phone can only ever create one kind of category — the type is hardcoded whatever the user picked — and the request succeeds, so the mistake is invisible until someone looks on the web and finds it in the wrong list.
D9	Mobile	The phone still reads a field the backend deleted, so anything that depended on it is quietly dead. Either move it onto whatever replaced it or remove the code that reads it along with everything it fed — and say which.
D10	Mobile	A stray character in the sensor details sent from lite mode ends up inside one of the stored values, so that sensor's details are wrong and its body part is left out of the session. A lite-mode recording should list the same body parts as the same recording in normal mode.

E · The system knows what went wrong and does not say

Every one of these is a support call. The information exists and is thrown away. E2 comes first — E1 and E3 are built on top of it.

E2 Anything a user gets wrong comes back as a server error HIGH BACKEND

Creating a recording, registering firmware and creating a company all report a duplicate value, a missing field and a genuine crash in exactly the same way, so no app can show anything useful.

DO

What the user can fix should say what to fix; only real failures should look like failures.

NOTE FOR YOU

The work-cycle import was the worst of these and is now fixed — it says which row and which column. Copy that shape to the rest. Agree the shape with the web developer before either side builds, or it gets built twice.

E1 Pages show a developer's note instead of the reason HIGH WEB

When a page fails to load, the customer is shown text a developer wrote for themselves — one screen literally reads “React query error thrown manually from ArchivedSessionsTable”. The real reason from the server is captured and never shown. There are twenty-seven of these in the code today.

DO

Build one way to turn a failed request into something on screen and use it everywhere: one place that reads the server's reason, one place that decides what the user sees.

Delete the ones on screens that no longer render rather than converting them — but do H2 first, or you will convert dead code.

NOTE FOR YOU

Roughly eight of the twenty-seven are still reachable: the manual assessment forms, the comparison page and the combined results pages. Depends on E2.

E4 The error screen can lock up the browser tab HIGH WEB

When the app catches an unexpected error it calls its own reset while it is still drawing the error screen, which causes the error again, which resets again. It spins until the tab freezes. Both apps.

DO

The error screen must not reset itself while rendering. Do it with E11 — same screen.

ID	WHERE	TASK
E3	Web	When the backend rejects a form and names the field, the client panel reads only the general message and shows a vague — sometimes empty — notification, with nothing marking the field. Put the message on the field it belongs to. The duplicate-email case now does this; nothing else does.
E6	Mobile / Web	Fields accept values the backend will refuse, so the user only finds out after submitting. On the phone: Worker ID, workplace, comment, age, weight, resting heart rate. On the web: company name, ticket subject, work-cycle step name. Take the limits from the backend's own model rather than inventing them.
E7	Web	Adding a factory with a name that already exists shows nothing on the field and the dialog just sits there. The team invite form also sends whatever is typed without checking it is a valid email address.
E8	Backend	Values that are too long reach the database and come back as an unreadable error.
E9	Backend	Updating several archive settings at once quietly skips any the user is not allowed to change and still reports success. Say what changed and what did not.

ID	WHERE	TASK
E10	Backend	Nothing stops two labels with the same name in one category, two operators with the same name on one workstation, or two report templates with the same name — so the user sees identical entries in a drop-down with no way to tell them apart.
E11	Web	“Try again” on the error screen only clears the error and draws the page again — it never asks the server for the data a second time, so the same error comes straight back. That screen is also hardcoded English in both apps: “Something went wrong” and “Try again” stay in English in every language.
E12	Mobile	The “no organisation” message is hardcoded English in two places — “Failed to create work session no Organization specified” — whatever the app language. Everything else is translated. Word it as something a user can act on rather than as a developer note.
E13	Backend	Updating firmware that does not exist reports a validation problem instead of “not found”.

F · Slow

The web items F1–F6, F8 and F9 are with the web team as *Web — speed cleanup* (17 August) and are not repeated here. What is left is the backend and the phone.

ID	WHERE	TASK
F7	Backend	Live-update features are wired to an in-memory channel layer, so they work in one server process only and fail silently the moment we run more than one.
F10	Mobile	The recordings list piles up background work as you scroll — each row starts several jobs and cleans up one, so a sync you started can keep running after you have left the screen. Everything a row starts must be cancelled when the row goes away.
F11	Mobile	The sensor clock is re-synced before every recording. There is a shortcut meant to skip it when it was done recently, and a units mistake means the condition can never be true, so it never fires and every recording starts slower than it needs to.
F12	Mobile	Switching between the two kinds of Bluetooth scan leaves the first one running. Starting the firmware scan cancels only the normal scan's listener and the reverse is also true, so the radio stays busy and stale devices show up in the new results.

G · Build, deploy and safety nets

G1 Broken code can be deployed DO FIRST WEB

The pipeline runs a type check, a linter, a build and the tests. Then it ships — from a step that never looks at whether any of them passed. The linter is explicitly told to carry on regardless, and so are both image builds, so a broken build and a good one look identical on the pull request.

DO

Make shipping depend on the checks passing. Nothing is published unless the type check, the linter, the build and the tests all succeeded. Do G2, G6 and G7 in the same job — they are the same pipeline.

We want to see each of the five failure cases go red before this is signed off.

NOTE FOR YOU

Top item on 10 July, on 27 July, on 12 August, still unchanged. Until this is done, anything fixed on the web side can come back the week after — and we have proof: two paid fixes were merged in May and silently undone by one merge in June, and nobody found out for eight weeks. The backend already has this. Copy what it does rather than designing something new.

ID	WHERE	TASK
G2	Web	High. The linter reports 430 problems, 52 of them errors, and the pipeline is told to carry on regardless so nobody sees them. Nearly half come from the admin app using a library it never declared — it works only because another app happens to install it. Clear them, declare the library, then let the linter stop a bad build. Clear first, gate second, or everyone is blocked.
G3	Web	High. The web apps have exactly two test files and both assert that true is true. Start with the RULA and RAMP scoring, the score-to-zone mapping and the score-to-colour mapping — the numbers that tell a customer whether their worker is at risk. Cover the boundary values specifically. The backend went from nothing to a full suite in about a week this month; point the web devs at what they did.

ID	WHERE	TASK
G4	Web	High. Neither app is given the backend's address properly when it is built, so any image built the normal way has no valid address — and yet production works, which means something undocumented is supplying it and nobody knows what. Find out and write it down first; that answer alone is worth the task. Then wire it up and fail the build loudly if it is missing.
G5	Backend	Half fixed. The housekeeping job now runs every five hours instead of every minute. It is still started once per server process with nothing stopping duplicates, so with more than one worker it still races against itself.
G6	Web	A broken image build keeps the pipeline green — both builds are told to carry on after a failure, so a broken build and a good one look identical on the pull request.
G7	Web	The admin build ignores the lockfile, so two builds of the same commit can install different software. The client panel already honours it.

H · Cleanup and dead ends

Nothing here affects a customer today. Most are traps for the next developer. H2 is the expensive one — it costs us time on every review.

H2 The old v1 screens are still in the codebase WEB

v2 was switched on everywhere in July. The v1 pages are still there, behind switches that can no longer be turned on, and a handful of old pages are still reachable by typing the address.

This is not tidiness. Most of the wrong findings in our last three audits came from someone reading a dead screen and reporting a bug no customer can reach.

DO

Confirm v2 is not being rolled back, then remove the v1 pages, the switches that used to select them, and the addresses that still reach them. Every address retired should be listed, so we can check none of them are in a customer’s bookmark. Do it after G1. Without a working gate, a removal this size is exactly the kind of change that breaks something quietly.

H11 One bad record from the server leaves the screen stuck loading MOBILE

A single malformed record crashes the whole work-cycle list instead of being skipped. The same happens on the company list and on email sign-in.

The shared safety net around repository calls only catches one of the two kinds of failure Dart has. A reply in an unexpected shape produces the other kind, so it goes straight past the net and the screen sits on its spinner forever.

DO

Widen the shared safety net first — it is one place and covers every screen. Then make the lists survive one bad row: skip it, show the rest, report the bad one to Sentry.

ID	WHERE	TASK
H1	Web	A developer debugging panel is rendered in production in both apps, and it is installed as a normal dependency rather than a development one, which is why nobody noticed.
H3	Web	Three places switch off type checking on data coming back from the backend, so a backend change breaks them for the user instead of failing the build. One of these caused the “Measured this month: 0” bug. Do it with H2.
H4	Web	Dead admin interface: a bulk archive action left commented out, a removed notifications bell whose button would not have worked, and a “?” icon that looks pressable and only has a tooltip. Do it with H2.
H5	Web	When a user exports a comparison they get a spreadsheet and a PDF that should share a name. The timestamp is worked out twice, so if the clock ticks between them the two files look unrelated — and around midnight they get different dates.
H6	Web	An unused file defines the “correct” names for cached assessment data while the whole app uses a misspelled version. Nothing is broken because everyone is wrong the same way — but the first developer who uses the correct file will silently break assessment refreshing, and it will look like an intermittent bug. Delete it or adopt it; do not leave both. Do it with H2.
H7	Backend	The work-cycle step list also returns steps belonging to no company. There are none in production and nothing in the product can create one, so this is a guard rather than a leak: return only the company’s own steps, and decide what happens to an orphan if one ever appears.
H8	Backend / Mobile	Worker records no longer have a first and last name, but both apps still carry them, the backend ignores them, and the phone splits the Worker ID into a pretend first and last name to display it. Nothing is lost — it is just confusing to read, and the next person here will assume those names mean something.
H9	Mobile	Switching company while offline reports success and never tells the server, so the user believes they are in the new company and their next recording is filed under the old one. Either refuse it offline and say why, or hold it

ID	WHERE	TASK
		and complete it when the connection returns — but do not show the user as switched until it has.
H10	Mobile	The first heart-rate reading after a pause measures its gap from before the pause, stretching it across the whole break and skewing the recovery figure.
H12	Web	Errors appear in the browser console on every page load in both apps. Harmless in themselves, but they bury real ones. Each is either a real problem or noise — fix or remove, do not silence, and tell us what each one turned out to be.
H13	Web	The prototype pages are open on the live site with no login. They touch no real data, but anyone with the address can look at them.
H14	Web	Plurals are hardcoded (“1 lines”, “1 stations”), gender shows raw and inconsistently — “male” next to “MALE” in one column — and times under a second show as 00:00:00 next to a percentage that is not zero.
H15	Web	Comparison columns say “Session 1 / Session 2” instead of the measurement numbers, and body parts that were not recorded show as full tables of dashes instead of being hidden.
H16	Web	The measurement page shows one “View RAMP” link even when several assessments exist, so duplicates are invisible from there.
H17	Web	The Home to-do count and the notification bell count what has been loaded rather than the real total, so both stop being right past a few hundred items.
H18	Web / Backend	The Workers list still counts records that were created automatically for staff before that behaviour was removed. New staff no longer get one, but the old rows are indistinguishable from real workers, so this is a data clean-up rather than a filter.

Decisions for you

Not developer calls. Nothing here should become a ticket until it is answered. DEC-1, DEC-7 and DEC-8 are closed — the head threshold, the AI fallback and the 1.5 colour rule are all settled and written.

DEC-6 · BLOCKS B2 AND B6

MEC question 11: what is it actually asking?

The screen calls it “neck bending”. The code works it out as the head angle minus the trunk angle. The backend’s own internal name for it says it should be the head’s own angle. Three names, one calculation.

With 12.1 and 12.2 being deleted, this is now a simple question: decide what question 11 measures. The title and the code then both follow from it. Two items are waiting on the answer.

DEC-9

Five illustration and range mismatches

1.2 head backward, 1.3 forward range, 1.3 side bending, 1.4 backward bending, 1.5 missing degrees. Each one needs a decision on which side changes — the illustration or the range — before any work starts. Assia has the list. The illustrations themselves are a conversation with Khadidja.

DEC-2

Do gender and dominant arm affect any calculation?

The corruption is fixed — a recording no longer overwrites what an admin set, and the phone now sends the values in a form the backend accepts. What is still unanswered is whether those two fields feed any calculation. If they do, results from before August were affected and we need to know how far back.

DEC-3

How much is section C worth to us?

Nothing in section C affects a customer and there is no sign any of it has been used. Fixing it properly is real work. The counter-argument is that we sell to several companies and one reading another’s data ends a deal and starts a GDPR conversation. Fund it now, or schedule it.

DEC-4

Should two companies be allowed the same name?

Company names are unique across the whole system, so the second company called “Nordic Logistics” to sign up is refused. This may be deliberate. Answer it on purpose rather than by accident.

DEC-5

Self-signup: build it or delete the page?

There is a “Create an Account” page with a button that does nothing, and nothing in the product links to it. Given that email sign-in and referral registration both shipped recently, the real question is whether self-signup is on the roadmap at all.

Features, not bugs

FEATURE

An assessment you can save and come back to

A RULA, REBA or RAMP form is all-or-nothing today: fill in half of it, click away, and everything since the last Save is gone with no warning. There is no autosave and no draft.

The answer is a proper unfinished state the user controls — save what you have, see it marked as unfinished in the list, open it later and carry on. Design it together with A2, which is the same screen.

FEATURE

Choose how many rows you see

No list in the product lets you change the number of rows per page. Measurements is fixed at ten. If you want a control for it, it is new work — the old one is gone with the screens it lived on.

Parked on purpose

Real findings, deliberately not scheduled. Listed so nobody re-reports them.

- **Location stops being recorded after the first session** — it works for the first recording after the app opens and never again. You chose to hold this.
- **TNO analysis drops a whole middle range of angles** — for the trunk that is 0° to 20°, most normal working posture. Farhad: skip TNO for now.
- **Phone warnings versus report thresholds for the head** — Farhad wants a table comparing the two before deciding.
- **Four invitation-flow flaws** — the admin invite path emails a dead link, a failed email locks the invite for seven days with no resend, member limits are stored but never enforced, and deleting a pending invitee can delete the person entirely. You asked to hold these.
- **Five sensor-configuration problems** — the one that matters is a recording being processed at the wrong sample rate when the phone does not report which it used, producing wrong durations and angle statistics with nothing flagging it. You asked for no tickets.

Closed since 12 August

Re-read in the code and confirmed done on `dev` . Listed so they are not re-reported.

ID	WHERE	TASK
A3	Web	Deactivated team members now have a Reactivate action in the same menu Deactivate sits in.
A4	Backend / Web	Changing an email to one already in use now returns “This email is already in use” and the client panel shows it on the field.
A6	All three	Work-cycle steps have a saved order, the drag handle sets it, the backend sorts by it and the phone follows it.
A8	Mobile	The server’s recording id is now written down before anything else can fail, so a sync no longer uploads a second copy.
A9	Backend	A work-cycle import is now all-or-nothing, and a bad row says which row and which column.
A10	Web	Work Library: the 5 MB limit is enforced, .csv/.xlsx/.xls are accepted properly, merging warns that it cannot be undone, and the retire warning shows how many tags go with it.
B1	Backend / Mobile	Head backward bending is -5° everywhere — risk verdict, Word report, RAMP and phone. Trunk stays -10° . This answers DEC-1.
B3	Backend / Web	A limb we did not measure no longer scores a green 0. A real zero shows as zero; a missing sensor shows as missing, with no score and no colour.
B5	Backend	Trunk and head forward/backward bending percentiles now print in the Word report. Side bending is left out, as decided.
B10	Web	RAMP item 1.5 asks its own question now, in all five languages.
B11	Web	RAMP sub-rows are named — forward, side and backward bending — instead of restarting at 1.1 inside every item.
B12	Backend / Web	Arm and forearm length are asked once per assessment and RAMP uses them, instead of assuming 32 cm and 27 cm for every worker.
D3	Mobile	Recorded body parts are sent in the form the backend stores, so they are no longer all reported as missing.
D6	Mobile	Choosing an operator now returns that operator’s work cycles instead of the whole company’s.
E5	Mobile	Opening the sensor scan with Bluetooth off now says so instead of spinning forever.

Written, waiting to be pushed

Done but not yet on `dev` , so they are not in the list above and were not re-read. Check them once they land.

- Score 1.5 is yellow on the auto-generated results page, matching the hand-filled page and the official RAMP guide. The duplicate colour rule is deleted and one rule covers both pages.
- Hand distance follows the selected task or timeframe. Question 1.6 was using the whole recording while 1.1 to 1.5 already did it properly.
- Hand distance is labelled as an estimate in the client panel. The Word report already said “estimated”.
- The head column is added to the per-task percentile table in the Word report. It only had back, left arm and right arm.
- MEC questions 12.1 and 12.2 are deleted. They were empty placeholders; nothing calculated or displayed them.
- The report no longer prints the raw AI error text under the “Summary” heading when the AI call fails. It prints the generic fallback sentence already in the code and sends an alert. This closes DEC-7.

How this was checked

- All four repositories were pulled and read on `dev` : backend `00c6e35` , web apps `cb2d905d` , mobile `96aed6a6` , landing `bb99528` (untouched since October 2024).
- Every commit made since the 12 August revision was read, and every item it could plausibly close was re-read in the code rather than taken from the commit message.
- Not re-read: A11 (admin forms), H3 to H6 and H12 to H18. These are small cleanup items that cannot be settled by reading code, and no commit has touched those files, so nothing can have changed. Nothing on the list depends on them.
- The web speed items are absent because they are already with the team as a separate sheet, not because they are done. The six calculation items above are absent because they are written but not yet pushed.